CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

Microtubule: 11 molecules stopped early

26 stopped trials across 11 molecules: Paclitaxel, Vincristine, Belantamab Mafodotin, Docetaxel Anhydrous and 7 more. 22 of the stops were the trial's own verdict and 1 followed a decision taken elsewhere; 3 cannot be established. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

4.0%
26 of 653 closed trials (95% CI 2.7%–5.8%)

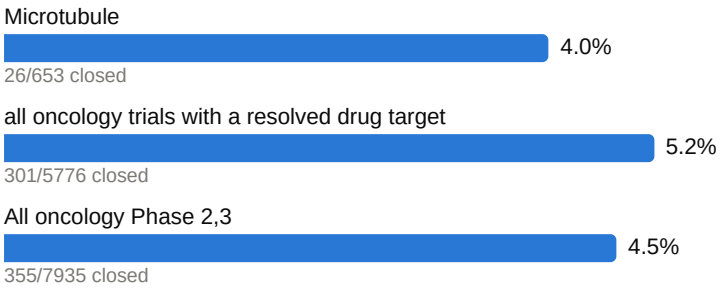
20 / 5 / 1
efficacy / safety / both (adds to 26)

34.6%
of 1885 trials have closed · 1232 still open or unresolved

1.5%
stopped for a biological reason within 3 years of starting (95% CI 0.9%–2.4%; 869 still at risk; 1.5% area-wide)

11 of 19 **Completed and missed the primary endpoint.** Of 451 completed trials in this segment, 19 posted a primary result on ClinicalTrials.gov that can be read — the sponsor's own comparison held to the sponsor's own threshold, or its statement that the endpoint was missed: 11 missed and 5 met, 3 split across co-primary endpoints. Counted separately and never in the rate above; a floor, not a rate. 194 of the 418 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 648 distinct experimental drugs across 1885 trials (small molecule 1612, antibody 902, adc 438, unknown 263). Most tested: PACLITAXEL (875), CARBOPLATIN (340), DOCETAXEL ANHYDROUS (313), CISPLATIN (240), CYCLOPHOSPHAMIDE ANHYDROUS (223). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 26 stops came from 26 sponsor-asset programmes across 23 sponsors; the largest (Wake Forest University Health Sciences) contributed 1; removing that programme's trials from both sides leaves 3.8%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 48 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 11.3%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	13	317	4.1%	2.4%–6.9%
2018–2020	6	190	3.2%	1.5%–6.7%
2021–2024	7	146	4.8%	2.3%–9.6%

Where the stops sit

Efficacy / futility: 20, Safety: 5, Efficacy and safety: 1.
Sponsors with most stops: GSK (3), Sanofi (2), Fox Chase Cancer Center (1), Spanish Breast Cancer Research Group (1), Tufts Medical Center (1), Jacob Soumerai, MD (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05420636	2	Fox Chase Cancer Center	IADADEMSTAT, PACLITAXEL	2025-07	Efficacy	Closed to accrual due to low probability of successful outcomes

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT05583110	2	Spanish Breast Cancer Research Group	TUCATINIB, TRASTUZUMAB, VINORELBINE	2025-05	Safety	The study was closed early due to safety concerns and an unfavorable benefit-risk balance. Unexpected high neutropenia rates required a costly sub-study, which was not feasible, forcing the sponsor to terminate the trial prematurely.
NCT04856176	2	Tufts Medical Center	Granulocyte-Macrophage Colony-Sti…, PEMBROLIZUMAB, PEMETREXED, PACLITAXEL	2024-12	Efficacy	Futility
NCT06964165	2	GSK GSK	NIRAPARIB, CARBOPLATIN, PACLITAXEL	2024-06	Efficacy	The study stopped after meeting pre-specified futility criteria
NCT06655818	2	GSK GSK	BELANTAMAB MAFODOTIN, DOSTARLIMAB	2024-02	Efficacy	The study was terminated due to lack of efficacy
NCT04676360	2	Jacob Soumerai, MD	BELANTAMAB MAFODOTIN	2023-11	Efficacy	The study was closed given lack of efficacy and plans to publish this negative study.

The 6 most recent of 26 stopped trials. Every molecule behind all 26 is named above; the complete trial list, the trials still running and the terminations with no readable cause ship with the dataset and the diligence report.

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 34.6% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. Free during the beta, in exchange for a work email, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.